

Program3

Code a Dockerized Python Flask or [Node.js](#) Application

Create a DevOps directory in that again create program3 directory

Within that create 3 files

requirements.txt

[app.py](#)

Dockerfile

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ vim requirements.txt
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ vim app.py
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ vim app.py
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ vim app.py
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ vim Dockerfile
```

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ cat app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from simlle Flask Docker"

if __name__ == "__main__":
    app.run(host="0.0.0.0" , port = 5000)
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ cat Dockerfile
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . /app/

EXPOSE 5000

CMD ["python" ,"app.py"]
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program3$ cat requirements.txt
Flask = 2.3.3
```

Now build a Dockerfile

sudo docker build -t program3 .

```

1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/Dev0ps/program3$ sudo docker build -t program3 .
[+] Building 12.2s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default 0.0s
=> => transferring dockerfile: 202B                                              0.0s
=> [internal] load metadata for docker.io/library/python:latest                 2.7s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 2B                                                  0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d 0.0s
=> [internal] load build context                                                0.0s
=> => transferring context: 118B                                               0.0s
=> CACHED [2/5] WORKDIR /app                                                    0.0s
=> [3/5] COPY requirements.txt .                                                0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt                    8.5s
=> [5/5] COPY . /app/                                                          0.2s
=> exporting to image                                                          0.4s
=> => exporting layers                                                         0.3s
=> => writing image sha256:c811b2d07efc8a5a5562cc26df098f67c36d77000182badd11008875763fef36 0.0s
=> => naming to docker.io/library/program3                                     0.0s

```

And now run docker
 sudo docker run -t 5000:5000 program3

```

1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/Dev0ps/program3$ sudo docker run -p 5000:5000 program3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
172.18.0.1 - - [07/Nov/2025 04:05:57] "GET / HTTP/1.1" 200 -
172.18.0.1 - - [07/Nov/2025 04:05:57] "GET /favicon.ico HTTP/1.1" 404 -

```

Output in the localhost



The screenshot shows a web browser window with the address bar set to 'localhost:5000'. The page content displays 'Hello from simlle Flask Docker'.